

Week- 1 Assignment

Part-1 : AI Concepts

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Title: AI Foundations Report

Q1. What is Artificial Intelligence (AI)?

Artificial Intelligence (AI) is a branch of computer science that enables machines to perform tasks that normally require human intelligence. AI systems can learn from data, recognize patterns, make decisions, and solve problems.

Key Capabilities of AI

- **Learning from Data:** AI improves its performance by analyzing large amounts of data.
- **Decision Making:** AI can make predictions and decisions based on patterns.
- **Language Understanding:** AI can understand and generate human language, as seen in ChatGPT and Gemini.

Applications of AI

- Google Maps for navigation and traffic prediction.
- YouTube and Netflix recommendation systems.
- Face recognition and virtual assistants.
- Medical diagnosis and self-driving vehicles.

Q2: Difference between AI, Machine Learning, Generative AI, and Agentic AI

Feature	AI	Machine Learning (ML)	Generative AI	Agentic AI
Definition	Technology that enables machines to perform intelligent tasks.	A subset of AI that learns from data without explicit programming.	AI that creates new content such as text, images, or code.	AI that can plan, make decisions, and complete tasks independently.
How It Works	Uses predefined rules, logic, and data to solve problems.	Learns patterns from data and improves over time.	Generates new content based on patterns learned from training data.	Breaks goals into steps, chooses actions, and works toward completing tasks.
Example	Chess-playing computer, virtual	Spam email filter, Netflix recommendations.	ChatGPT, DALL·E, Midjourney.	AI travel assistant, autonomous task

Feature	AI	Machine Learning (ML)	Generative AI	Agentic AI
	assistant.			manager.

Q3: Explain any three open-source AI models available on Hugging Face

- **DeepSeek-R1**
 - **Primary Use Case: Advanced Reasoning** – Used for solving difficult math, deep logic, and complex coding problems.
 - **Key Feature: Thinking Mode** – It spends extra time thinking step-by-step internally before giving the final answer.
 - **Example:** Helping a student debug complex Python loops or solving hard mathematical theorems.
- **Llama 3**
 - **Primary Use Case: Text Generation & Chat** – Used for writing essays, summarizing text, and answering general knowledge questions.
 - **Key Feature: Smart Conversations** – It is highly optimized to understand and chat naturally like a human.
 - **Example:** Asking it to write a formal email or summarize a huge academic paragraph into simple points.
- **Stable Diffusion**
 - **Primary Use Case: Image Generation** – Used for creating high-quality digital art, logos, and realistic photos from text descriptions.
 - **Key Feature: Text-to-Image Creation** – It can understand descriptive words and instantly turn them into high-resolution visuals.
 - **Example:** Typing "A cute robot coding on a laptop" and getting a high-quality picture instantly.

Q4: Difference between Traditional Programming and AI Systems

S.No	Main Point	Traditional Programming	AI Systems (Machine Learning)
1	Concept	Developers write all rules manually.	System learns patterns from data.
2	Input & Output	Data + Rules = Output	Data + Output = Learned Rules
3	Handling New Data	May fail on unfamiliar situations.	Can make predictions on new data.
4	Human Effort	More coding effort is required.	More data collection and training effort is required.

S.No	Main Point	Traditional Programming	AI Systems (Machine Learning)
5	Example	Calculator application	Spam email filter

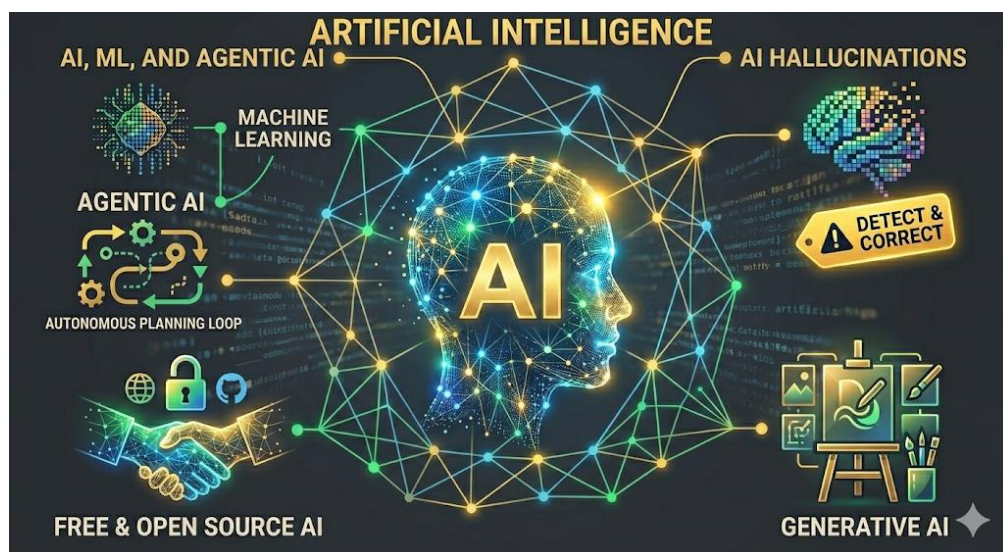
Q5: What is AI Hallucination?

Definition: AI Hallucination means a situation where an AI model generates completely false, incorrect, or unrealistic information, but presents it with full confidence as if it is a true fact.

Why It Happens: It happens because AI does not actually "know" the truth like humans; it only guesses the next most matching words based on past data patterns. If the data is missing or confusing, the AI just makes up a fake story.

How to Fix It: Users can reduce hallucinations by giving very clear instructions (prompts), asking the AI to double-check its source, or telling it to say *"I don't know the answer"* if it is not sure.

Example: If you ask an AI to write a biography about a normal person, it might confidently invent fake awards, college degrees, or job history that the person never actually achieved.



ARTIFICIAL INTELLIGENCE

